

Visual Forensics Audit

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1 Scope

This report summarizes text-driven visual-forensics heuristics from figure-caption extraction. It also reports pilot human-in-the-loop plot digitization coverage.

2 Load Outputs

```
repo_root <- normalizePath("../", winslash = "/", mustWork = TRUE)
study_id <- Sys.getenv("FORENSICS_STUDY_ID", "lungtime")
report_dir <- file.path(repo_root, "reports", "visual", study_id)

required <- c("visual_row_results.csv", "visual_summary.csv", "visual_digitized_values.csv")
missing <- required[!file.exists(file.path(report_dir, required))]
if (length(missing) > 0) {
  stop("Missing files: ", paste(missing, collapse = ", "))
}
```

```

row_results <- read.csv(file.path(report_dir, "visual_row_results.csv"), check.names = FALSE)
summary_tbl <- read.csv(file.path(report_dir, "visual_summary.csv"), check.names = FALSE)
digitized <- read.csv(file.path(report_dir, "visual_digitized_values.csv"), check.names = FALSE)

compact_character <- function(x, width = 44) {
  x <- as.character(x)
  x[is.na(x)] <- ""
  too_long <- nchar(x) > width
  x[too_long] <- paste0(substr(x[too_long], 1, width - 3), "...")
  x
}

compact_df <- function(df, cols = NULL, n = 25, char_width = 44) {
  out <- df
  if (!is.null(cols)) {
    keep <- intersect(cols, names(out))
    out <- out[, keep, drop = FALSE]
  }
  if (nrow(out) > n) {
    out <- utils::head(out, n)
  }
  is_char_like <- vapply(out, function(col) is.character(col) || is.factor(col), logical(1))
  out[is_char_like] <- lapply(out[is_char_like], compact_character, width = char_width)
  out
}

```

3 Interpretation Key

- `n_figure_mentions`: number of caption-like figure mentions extracted from PDF text.
- `n_duplicate_pairs`: number of near-duplicate caption pairs above similarity threshold.
- `near_duplicate_rate`: duplicate-pair burden relative to extracted figure mentions.
- `n_numbering_gaps`: count of missing figure numbers between observed min and max.
- `numbering_gap_flag`: quick signal that sequence gaps exist in extracted numbering.
- `n_digitized_figures`: unique figures with digitized point data.
- `n_digitized_series`: unique digitized series across included figures.
- `n_digitized_points`: total extracted points available for downstream analyses.
- `digitization_ready`: whether non-empty digitized data are available in this run.

Digitization values come from manual `metaDigitise` calibration/click workflows and should be treated as measurement data with operator uncertainty.

```
knitr::kable(summary_tbl, digits = 4)
```

trial_id	n_figure_men- tions	n_duplicate_pairs	near_duplicate_rate	n_number-ing_gaps	num-ber-sing_gap_flag	n_digitized_fig- ures	n_digitized_se- ries	n_digitized_point- s	digitiza- tion_ready
pronto	7	0	0	0	FALSE	0	0	0	FALSE

4 Plot Digitization Coverage

```
knitr::kable(summary_tbl[, c(
  "trial_id",
  "n_digitized_figures",
  "n_digitized_series",
  "n_digitized_points",
  "digitization_ready"
)], digits = 4)
```

trial_id	n_digitized_figures	n_digitized_series	n_digitized_points	digitization_ready
pronto	0	0	0	FALSE

```
if (nrow(digitized) > 0) {
  knitr::kable(
    compact_df(
      digitized,
      cols = c(
        "study_id", "figure_id", "panel_id", "series_id",
        "x_value", "y_value", "x_unit", "y_unit",
        "extract_confidence", "source_image"
      ),
      n = 25,
      char_width = 38
    ),
    digits = 4
  )
} else {
  cat("No digitized plot values available yet (pilot capture not completed).")
}
```

No digitized plot values available yet (pilot capture not completed).

5 Row-Level Checks

```
knitr::kable(  
  compact_df(  
    row_results,  
    cols = c(  
      "trial_id",  
      "n_figure_mentions",  
      "n_duplicate_pairs",  
      "n_numbering_gaps",  
      "numbering_gaps",  
      "n_digitized_figures",  
      "n_digitized_series",  
      "n_digitized_points",  
      "digitization_ready"  
    ),  
    n = 20,  
    char_width = 36  
  )  
)
```

trial_id	n_fig- ure_men- tions	n_dupli- cate_pairs	n_num- ber- ing_gaps	num- ber- ing_gaps	n_digi- tized_fig- ures	n_digi- tized_se- ries	n_digi- tized_point	digitiza- tion_ready
pronto	7	0	0	NA	0	0	0	FALSE